

Homework 2

Due Date: 2 p.m., April 27 (Mon)

1. Consider the following problem.

$$\begin{array}{ll}\max & Z = x_1 + 2x_2, \\ \text{s.t.} & x_1 \leq 2 \\ & x_2 \leq 2 \\ & x_1 + x_2 \leq 3 \\ & x_1 \geq 0, \quad x_2 \geq 0.\end{array}$$

- (a) Plot the feasible region and circle all the CPF solutions.
- (b) For each CPF solution, identify the pair of constraint boundary equations that it satisfies.
- (c) For each CPF solution, use this pair of constraint boundary equations to solve algebraically for the values of x_1 and x_2 at the corner point.
- (d) For each CPF solution, identify its adjacent CPF solutions.
- (e) For each pair of adjacent CPF solutions, identify the constraint boundary they share by giving its equation.

2. Consider the following linear programming model.

$$\begin{array}{ll}\max & Z = x_1 + 2x_2, \\ \text{s.t.} & x_1 + 3x_2 \leq 8 \\ & x_1 + x_2 \leq 4 \\ & x_1 \geq 0, \quad x_2 \geq 0.\end{array}$$

- (a) Use graphical analysis to identify all the corner-point solutions for this model. Label each as either feasible or infeasible.
- (b) Calculate the value of the objective function for each of the CPF solutions. Use this information to identify an optimal solution.
- (c) Use the solution concepts of the simplex method to identify which sequence of CPF solutions might be examined by the simplex method to reach an optimal solution.

3. Work through the simplex method step by step (in tabular form) to solve the following problem.

$$\begin{array}{ll}\max & Z = 2x_1 + 4x_2 + 3x_3, \\ \text{s.t.} & 3x_1 + 4x_2 + 2x_3 \leq 60 \\ & 2x_1 + x_2 + 2x_3 \leq 40 \\ & x_1 + 3x_2 + 2x_3 \leq 80 \\ & x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.\end{array}$$

4. Consider the following problem.

$$\begin{array}{ll}\max & Z = 4x_1 + 2x_2 + 3x_3 + 5x_4, \\ \text{s.t.} & 2x_1 + 3x_2 + 4x_3 + 2x_4 = 300 \\ & 8x_1 + x_2 + x_3 + 5x_4 = 300 \\ & x_j \geq 0, \quad \forall j = 1, 2, 3, 4.\end{array}$$

(a) Using the Big M method to solve the problem step by step.

(b) Using the two-phase method to solve the problem step by step.

5. Consider the following problem.

$$\begin{array}{ll}\max & Z = -x_1 + 4x_2, \\ \text{s.t.} & -3x_1 + x_2 \leq 6 \\ & x_1 + 2x_2 \leq 4 \\ & x_2 \geq -3 \\ & \text{no lower bound constraint for } x_1.\end{array}$$

(a) Solve this problem graphically.

(b) Reformulate this problem so that it has only two functional constraints and all variables have nonnegativity constraints.

(c) Work through the simplex method step by step to solve the problem.